RFM Segmentation Report

UK Online Retail — Customer Scoring, Segmentation & Campaign Strategy

5,848	£16.9M	57.8%	77.2%
TOTAL CUSTOMERS	TOTAL REVENUE	CHAMPION REVENUE SHARE	TOP 20% REVENUE SHARE

DATASET	UCI Online Retail II — UK e-commerce transactions
PERIOD	December 2009 – December 2011
RAW DATA	~1,067,371 rows → ~824,000 rows after cleaning
TOOLSTACK	Python (Pandas, Plotly, SQLAlchemy) + MySQL 8.0
TECHNIQUE	RFM Scoring via NTILE(4) window functions

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1. Executive Summary

5,848	57.8%	£2.2M	77.2%
TOTAL CUSTOMERS	CHAMPION REVENUE	REVENUE AT RISK	TOP 20% REVENUE SHARE

5,848 customers of a UK online retailer were scored on three dimensions: how recently they bought, how often, and how much they spent. From that, they were split into 11 segments. The goal was straightforward. Figure out who the best customers are, who is slipping away, and what to do differently for each group.

1,008 customers sit in the Champions segment and account for 57.8% of all revenue. The top 20% of customers by spend generate 77.2% of total revenue. At Risk and About to Sleep customers together hold £2.2M of spend that has gone quiet. Every customer now has a score, a segment label, and a suggested action.

Metric	Finding
Total customers segmented	5,848
Total revenue (dataset)	£16.9M
Champions (customers / revenue)	1,008 customers → 57.8% of revenue
At Risk + About to Sleep revenue	£2,218,964
80/20 finding	Top 20% of customers → 77.2% of revenue
Cannot Lose Them	0 customers qualified in this dataset
Highest AOV segment	Lost (£685.61 avg order value)

2. Project Setup & Objectives

The Problem

Most online retailers send the same email to everyone. A win-back discount goes to someone who bought last week. A customer who has spent £9,000 across 17 orders gets the same newsletter as someone who bought once in 2010. RFM scoring addresses that. Each customer gets a score on three dimensions, then gets grouped with others who behave the same way.

What RFM Measures

Dimension	Question It Answers	Scoring Direction
Recency	How recently did they last buy?	Lower days = higher score
Frequency	How many times have they bought?	More orders = higher score
Monetary	How much have they spent total?	Higher spend = higher score

SQL Architecture

Three views were built in MySQL, each sitting on top of the previous one. This was intentional. Each layer can be checked on its own before moving to the next, which makes debugging much easier.

View	What It Does
v_rfm_raw	Aggregates Recency, Frequency, Monetary per customer. Reference date = MAX(invoicedate) + 1 day to simulate "today" on historical data.
v_rfm_scored	Applies NTILE(4) window functions to assign scores 1–4 on each dimension. Score 4 = best.
v_rfm_segments	CASE WHEN logic maps score combinations to 11 business segment labels.

3. Dataset & Data Cleaning

Dataset

Detail	Value
Source	UCI Machine Learning Repository — Online Retail II
Raw rows	~1,067,371 transactions
Rows after cleaning	~824,000 rows
Unique customers	5,848
Date range	December 2009 – December 2011
Region	UK-based online retailer (international customers included)
Load method	CSV → cleaned in Python → loaded to MySQL via pandas.to_sql()

Cleaning Steps

Nine steps were applied in sequence. Each one is documented with a reason. Dropping rows without explaining why makes the analysis hard to reproduce or audit later.

Step	Action	Reason
1	Standardised column names	Lowercase + underscores for consistent Python/SQL access
2	Removed negative quantities	Cancellations appear as negative qty in this dataset, not C-prefix invoices
3	Removed non-product stock codes	POST, DOT, BANK CHARGES etc. are operational entries, not real sales
4	Dropped missing Customer IDs	RFM needs a customer identifier — no ID means no segment
5	Removed zero-price items	Zero price = free sample or data error, not a real purchase
6	Fixed data types	customer_id cast to int, invoicedate cast to datetime
7	Capped price outliers (99.9p)	Extreme bulk orders distort monetary scores for normal customers
8	Created total_revenue column	quantity × price, rounded to 2dp — needed for MySQL aggregations
9	Removed exact duplicates	Final deduplication check

4. RFM Methodology

How NTILE Scoring Works

Each customer gets a score of 1 to 4 on each RFM dimension using NTILE(4). NTILE splits the ranked customer list into four equal-sized buckets. Score 4 is always the best: most recent buyer, most frequent buyer, or highest spender.

> **Why NTILE instead of fixed thresholds?** A fixed threshold like "score 4 if spent over £500" hardcodes assumptions about this specific dataset. Apply it to a different retailer or a different time period and the numbers stop making sense. NTILE scores customers relative to each other, not against an absolute number, so it works on any dataset without adjustment.

Dimension	NTILE ORDER BY	Score 1 (worst)	Score 4 (best)
Recency	recency_days DESC	Bought a very long time ago	Bought very recently
Frequency	frequency ASC	Bought very rarely	Buys very often
Monetary	monetary ASC	Spends very little	Spends the most

The 11 Segments

Score combinations map to segment labels through a CASE WHEN block. The conditions run in priority order, so the first one that matches wins.

Segment	Score Logic	Business Meaning
Champions	R=4, F>=3, M>=3	Best customers: recent, frequent, high spend
Loyal Customers	R>=3, F>=3	Buy often and recently, solid relationship
Potential Loyalists	R>=3, F>=2	Recently active, starting to show loyalty
New Customers	R=4, F=1	Bought recently but only once
Promising	R=3, F=1	Recent first-timers, worth nurturing
Need Attention	R<=2, F<=2, M<=2	Scores dropping across all three dimensions
About to Sleep	R=2, F<=2, M<=2	Low activity recently, borderline churn
At Risk	R<=2, F>=3, M>=3	Used to buy regularly and spend well, now quiet
Cannot Lose Them	R=1, F>=3, M>=3	Longest lapse, but historically very valuable
Hibernating	R=1, F>=2	Long-lapsed, has some purchase history
Lost	All others	No clear signal on any dimension

5. Segment Results

5.1 Customer Count by Segment

Loyal Customers (1,093) and Champions (1,008) together account for 36% of all customers. Need Attention has the most customers of any declining segment at 807. New Customers is the smallest group at 80, which means very few people made their first purchase in the most recent window.

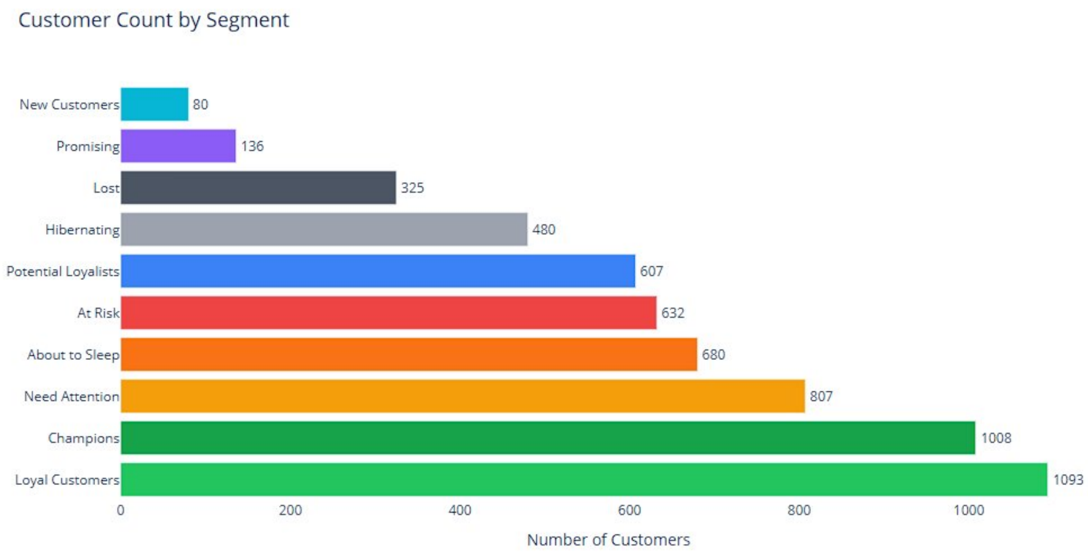


Figure 1: Customer count per segment

5.2 Revenue by Segment

Once you look at revenue instead of headcount, the picture shifts completely. Champions: £9.8M. Loyal Customers: £3.3M. All other 9 segments combined: roughly £3.8M. Two groups out of eleven account for 77% of what this business makes.

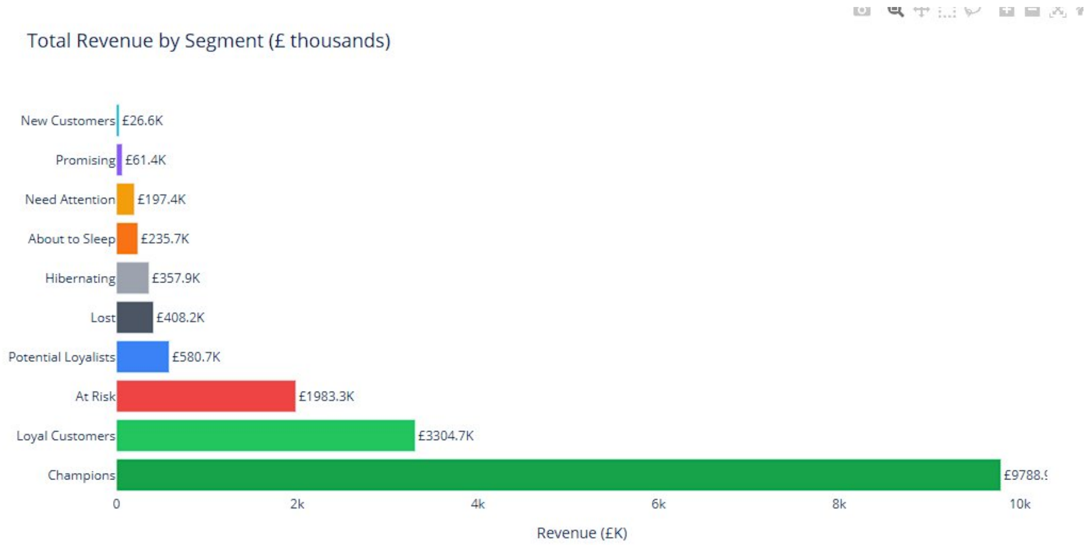


Figure 2: Total revenue per segment (£ thousands)

5.3 RFM Scores Heatmap

Each cell shows the average score (1 to 4) for that segment on that dimension. Champions come in at 4.0 on Recency, 3.72 on Frequency, and 3.70 on Monetary. The At Risk row is the one worth studying closely. Recency sits at 1.78, meaning they have not bought in a long time. But Frequency is 3.40 and Monetary is 3.38, so these were genuinely good customers. They just stopped coming back. New Customers score 4 on Recency but 1 on Frequency. One purchase, nothing more.

Figure 3: Average RFM scores by segment (1 = worst, 4 = best)

5.4 Recency vs Lifetime Value

Champions and Loyal Customers pile up in the bottom-left corner: recent buyers with high total spend. The red At Risk dots are scattered far to the right, 100 to over 700 days along the x-axis, but sitting high on the y-axis. Big spenders who have not returned. Those are the customers a win-back campaign should focus on.

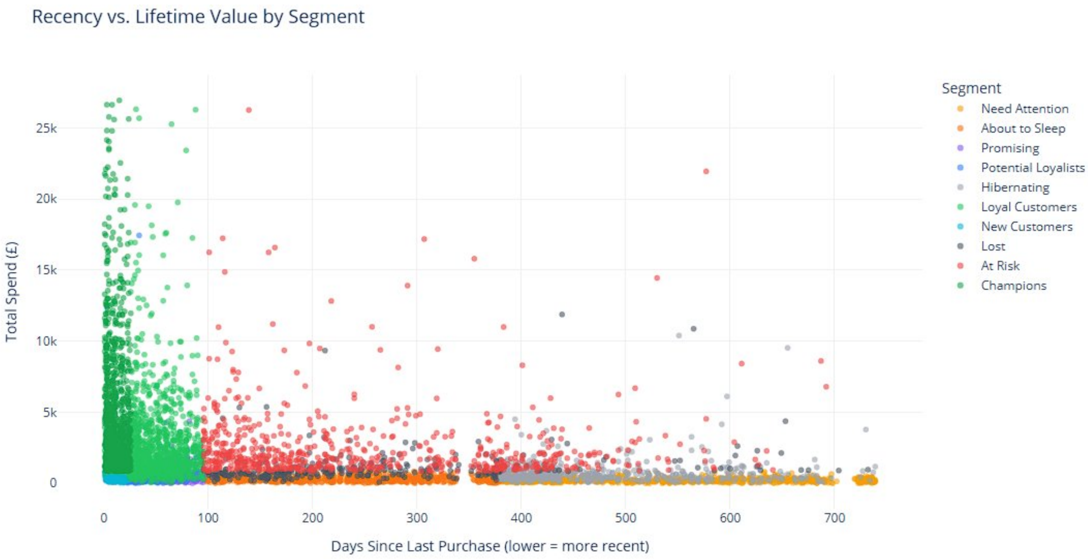


Figure 4: Recency vs. lifetime value per customer

6. Business Questions

Q1 — Who are your Champions, and what do they contribute?

1,008	£9.8M	57.8%	17.8
CHAMPIONS	TOTAL REVENUE	% OF ALL REVENUE	AVG ORDERS

1,008 customers. 57.8% of all revenue. The average Champion has placed 17.8 orders, spent £9,711 total, and last bought 11 days ago. These are not customers to win back. They are customers to keep. Any budget going toward acquiring new customers should come after this group is looked after.

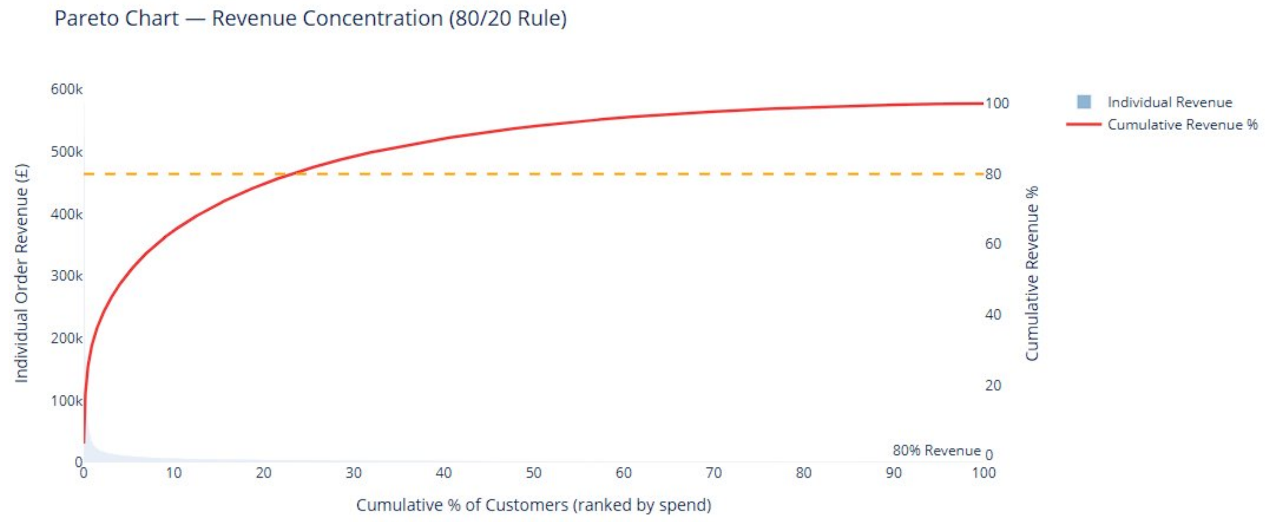
Q2 — Which customers are at risk, and how much revenue is at stake?

Segment	Customers	Revenue at Risk	Avg Days Since Purchase
At Risk	632	£1,983,268	253 days
About to Sleep	680	£235,696	244 days
Total	1,312	£2,218,964	—

At Risk customers scored 3.40 on Frequency and 3.38 on Monetary. They used to buy regularly and spend well. The average gap since their last purchase is 253 days, around 8 months. A win-back campaign targeting this group has more revenue upside than any other option right now.

Q3 — Does the 80/20 rule apply?

The top 20% of customers account for 77.2% of total revenue. The top 40% reach 90.0%. The bottom 20% contribute 1.1%. The strict 80/20 rule does not quite hold here, it is closer to 77/20. But the point is the same: a small group of customers is doing almost all the work.

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80/20 FINDING:
Top 20% of customers generate 77.2% of total revenue.
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Figure 5: Pareto chart — cumulative revenue % (top 20% → 77.2%)

80/20 FINDING

Top 20% of customers generate 77.2% of total revenue. Top 40% → 90.0%. Bottom 20% → 1.1%.

Customer Group	Revenue %	Cumulative Revenue %
Top 20% (Quintile 5)	~57%	77.2%
Upper-Mid (Quintile 4)	~13%	90.0%
Mid (Quintile 3)	~7%	~97%
Lower-Mid (Quintile 2)	~2%	~99%
Bottom 20% (Quintile 1)	1.1%	100%

Q4 — Full segment breakdown

Segment	Customers	% Customers	Total Revenue	% Revenue	Avg AOV
Champions	1,008	17.2%	£9,788,939	57.8%	£433.93
Loyal Customers	1,093	18.7%	£3,304,700	19.5%	£360.73
At Risk	632	10.8%	£1,983,268	11.7%	£407.97
Potential Loyalists	607	10.4%	£580,700	3.4%	£491.83
Lost	325	5.6%	£408,200	2.4%	£685.61
Hibernating	480	8.2%	£357,900	2.1%	£313.41
About to Sleep	680	11.6%	£235,700	1.4%	£229.70
Need Attention	807	13.8%	£197,400	1.2%	£244.62
Promising	136	2.3%	£61,400	0.4%	£451.13
New Customers	80	1.4%	£26,600	0.2%	£331.90
Cannot Lose Them	0	0.0%	—	—	—

Q5 — How does Average Order Value differ across segments?

AOV here is total spend divided by number of orders per customer. The Lost segment comes out on top at £685.61 per order. These customers did not buy often, but when they did, the orders were large. That raises a question worth investigating: did they leave because of a one-off purchase need, a pricing issue, or a bad experience? Champions sit fourth at £433.93. Their revenue lead comes from buying 17.8 times on average, not from unusually large orders.

7. Campaign Strategy

Below is the recommended action for each segment. Ordered by revenue, not by how many customers are in each group.

Customer Segment Strategy Recommendations

Segment	Customers	% Revenue	Action	Channel
Champions	1008	57.8	Reward & Retain	Personalized email, exclusive events
Loyal Customers	1093	19.5	Upsell & Cross-sell	Email, targeted ads
Potential Loyalists	607	3.4	Nurture to Loyalty	Email, push notifications
New Customers	80	0.2	Onboard & Engage	Welcome email sequence
Promising	136	0.4	Build Relationship	Email, retargeting ads
Need Attention	807	1.2	Re-engage	Email with discount
About to Sleep	680	1.4	Reactivate Now	Urgent email + SMS
At Risk	632	11.7	Win Back	Email + personal call for top spenders
Cannot Lose Them	null	null	Win Back Immediately	Personal phone call + email
Hibernating	480	2.1	Low-cost Re-engagement	Low-frequency email
Lost	325	2.4	Final Attempt or Suppress	Single email

Figure 6: Campaign strategy table from Python output

Segment	Priority	Action	Channel
Champions	Critical	Reward & Retain	VIP email, exclusive events, referral programme
Loyal Customers	High	Upsell & Cross-sell	Email, targeted ads, bundle offers
At Risk	Urgent	Win Back	Strong reactivation offer + personal call for top spenders
Potential Loyalists	High	Nurture to Loyalty	Membership programme invite, personalised recommendations
About to Sleep	Urgent	Reactivate Now	Time-limited discount + SMS
Need Attention	Medium	Re-engage	Email with discount, highlight new arrivals
New Customers	Medium	Onboard & Engage	Welcome sequence, product discovery, 2nd-purchase incentive
Promising	Medium	Build Relationship	Showcase popular products, limited-time offer
Hibernating	Low	Low-cost Re-engagement	Occasional newsletter, low-frequency email
Lost	Minimal	Final Attempt or Suppress	Single win-back email, then remove from active lists
Cannot Lose Them	N/A	No customers in dataset	If/when appears: executive outreach + best offer

> **Cannot Lose Them had zero customers in this dataset.** The condition requires R=1 (longest lapse) combined with F>=3 and M>=3 (historically frequent and high-spending). Nobody in this dataset hit all three simultaneously. In a live rolling dataset this segment would show up, and when it does it should be the first group contacted.

8. Conclusion

1,008 customers generate 57.8% of this business's revenue. That is the starting point for any budget conversation. The At Risk group (632 customers, £1.98M) is the clearest immediate opportunity. They bought regularly, spent well, and have not returned in an average of 253 days. A win-back campaign targeting this group has better ROI than anything else available.

The Pareto analysis puts the split at 77/20, not quite 80/20 but close enough to matter. Losing one Champion is not the same as losing one Need Attention customer. The numbers make that obvious, and marketing spend should reflect it.

Lost customers have the highest average order value in the dataset at £685.61, higher than Champions. These were not casual buyers who drifted away. They placed large orders, then stopped. Figuring out why could open up a recovery angle the current strategy does not address.

Metric	Result
Customers segmented	5,848
Total revenue covered	£16.9M
Revenue in top 2 segments	£13.1M (77.3% of total)
Revenue at risk	£2.2M (At Risk + About to Sleep)
Highest AOV segment	Lost — £685.61
Top 20% customers → revenue	77.2%
Cannot Lose Them customers	0 in this dataset
Recommended immediate action	Win-back campaign targeting At Risk segment